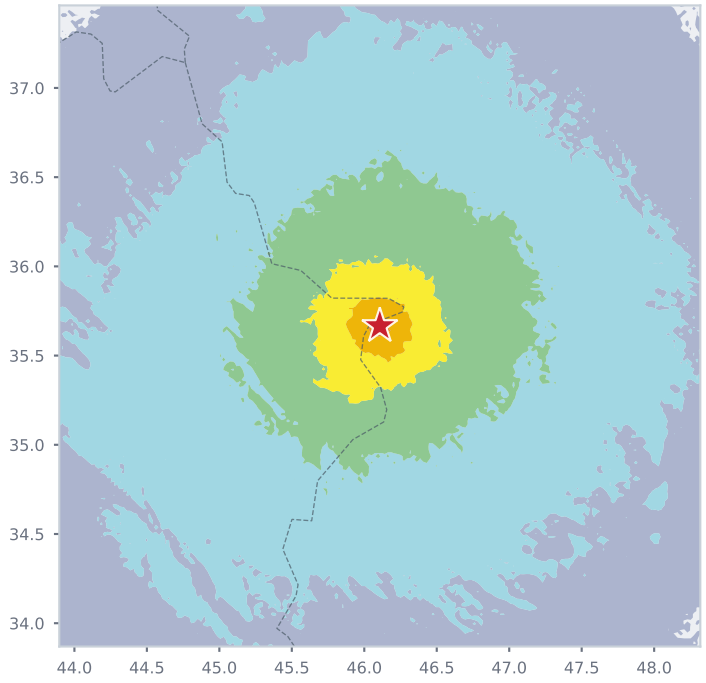
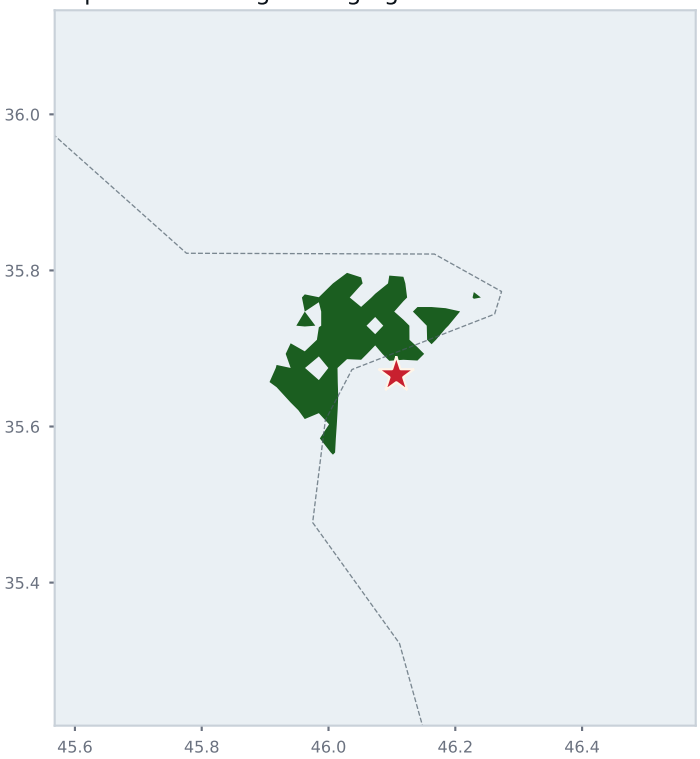
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity



Expected damage grade (IMS-25)



Damage alert

ORANGE Widespread damage; many buildings unusable **Provisional** **Night occupancy**

Buildings heavily damaged (DG3 or worse)

570 to 7.1 thousand

Central estimate 1.8 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

2.3 million

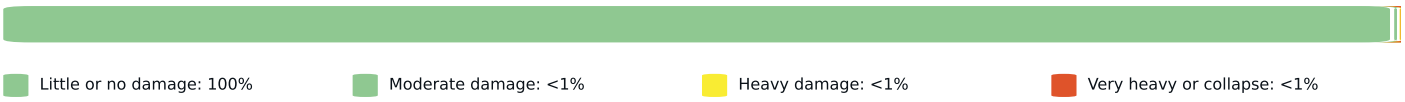
21% of them inferred from built-up area, not observed footprints.

People by shaking level

VIII	810
VII	46 thousand
VI	191 thousand
V	1.3 million
IV	4.6 million

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



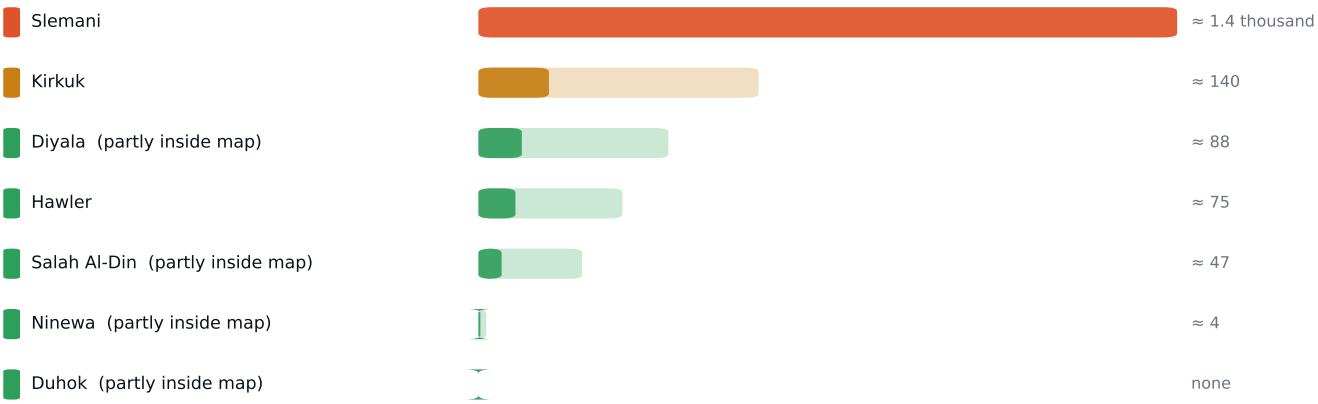
Little or no damage: 100% | Moderate damage: <1% | Heavy damage: <1% | Very heavy or collapse: <1%

Which buildings failed

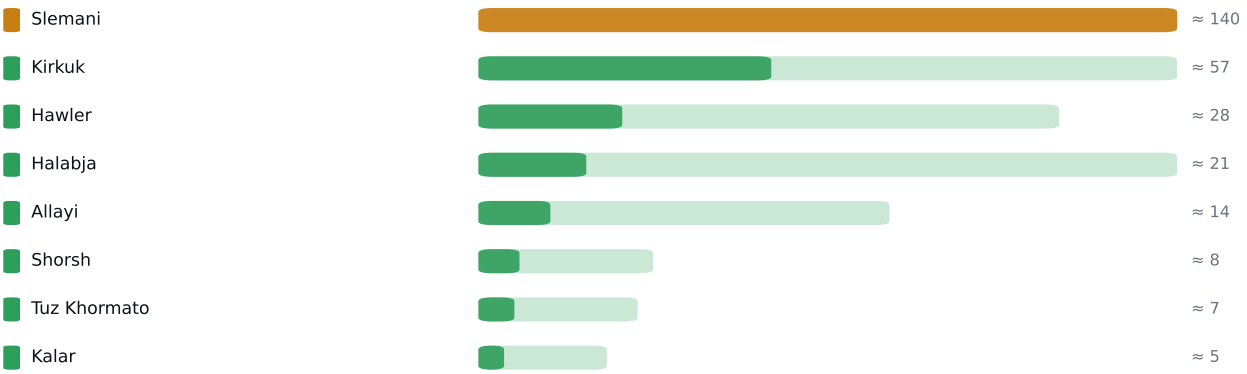
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**0.3%**	≈ 200 buildings
Adobe, earth brick	class A	**0.2%**	≈ 160 buildings
Manufactured stone units, timber floors	class B	**0.1%**	≈ 230 buildings
Simple stone	class B	**0.1%**	≈ 140 buildings
RC flat slab and others	class B	**0.1%**	≈ 1 buildings
Manufactured stone with RC floors	class C	**0.0%**	≈ 380 buildings
RC frame — no / low code (ERD-L)	class C	**0.0%**	≈ 41 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Slemani	669 thousand	950	420 to 5.4 thousand	0.14%
Kirkuk	459 thousand	96	46 to 570	0.02%
Hawler	568 thousand	59	25 to 290	0.01%
Diyala	300 thousand	59	27 to 380	0.02%
Salah Al-Din	243 thousand	32	14 to 210	0.01%
Ninewa	33 thousand	3	1 to 16	0.01%
Duhok	320	none	none to none	0%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 0 felt-report cells (felt channel: none)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml19460001, version 2, generated 19 Oct 2026, 11:32 UTC.